

JIGS AND FIXTURES TOOL DESIGN

MEM 425 HAND II MANUFACTURING OPTIMAL

1.0. GENERAL INTRODUCTION:

Jigs and fixture are special purpose tools which are used to facilitate production (machining, assembling and inspection operations) when workpieces are to be produced on a mass scale. The mass production of workpiece is based on the concept of interchangeability according to which every part will be produced within an established tolerance.

Jigs and fixtures provide a means of manufacturing interchangeable parts since they establish a relation, with predetermined tolerance, between the work and the cutting tool. They eliminate the necessity of a special set up for each individual part. Once a jig or fixture is properly set up, any number of duplicate parts may be readily produced without additional set up.

1.1 USES OF JIGS AND FIXTURES.

1. To reduce the cost of production.
Use eliminate the ~~losing~~ ^{setting up} out of work and setting up of tools.
2. To increase the production.

3. To assure high accuracy of the parts.
4. To provide for interchangeability.
5. To enable heavy and complex shaped parts to be machined by being held rigidly to a machine.
6. Reduced quality control expenses.
7. Increased versatility of machine tool
8. Less skilled labour
9. Saving labour
10. They use partially automates the Machine tool
11. They use improves the safety at work, thereby lowering the rate of accidents.

A Jig may be defined as a device which holds and positions the work, locates or guides the cutting tool relative to the workpiece and usually is not fixed to the machine table. It is usually lighter in construction.

A Fixture is a work holding device which only holds and positions the work, but does not in itself guide, locate or position the cutting tool. The setting of the tool is done by machine adjustment and a

Setting block or by using Slip gauge. A fixture is bolted or clamped to the machine table. It is usually heavy in construction.

1.2 AREA OF APPLICATIONS.

Jigs are used on drilling, Reaming, Tapping and Counter boring operations.

Fixtures are used in connection with turning, Milling, grinding, Shapping, Planing and boring operations.

Jigs and fixture, because of their functions and advantages are also called production devices. To facilitate interchangeability we need inspection devices, that is different types of gauges.

1.3 FUNCTIONS OF JIGS AND FIXTURES.

To fulfil their basic functions, both jigs and fixtures should possess the following components or elements.

1. A Sufficiently rigid body (Plate, box or frame structure) into which the workpiece are located.

2. Locating element

3. Clamping element

- 4 Tool guide element (for jig) or tool setting element (for fixture)
5. Element for positioning or fastening the jig or fixture on the machine on which it is used.

According to the degree of Mechanization and automation, Jigs and fixtures are classified as

- (a) Hand operated
- (b) Power operated
- (c) Semi-automatic
- (d) Automatic.

MATERIALS FOR JIGS AND FIXTURES

1. Cast Iron, because of its damping capacity it is a very important material for the manufacture of jigs and fixture. For heavy bodies, it is always economical to use cast iron. Also Complicated and detailed machining can be avoided. Where lightness is the main consideration.

2. Alloys of aluminium and magnesium. In the case of screw operated feeding and clamping devices,

3. Phosphor bronze replaceable nuts are used, screws being costly items. Nuts will wear out early and can be replaced. They reduce the wear of the screws and also provide corrosion resistance.

JIGS AND FIXTURES CONSTRUCTION.

The body of a jig and a fixture can be constructed by any of the following methods.

① Casting

2. Fabrication

3. Welding

In the last two methods, the body is made up from separate parts.

The advantages of the Cast Construction are

- i) Jigs and fixture with complicated shapes can be easily cast

- ii) Cast iron has the property of absorbing and damping out the vibrations.

- iii) Any number of castings with the same characteristics can be made from one pattern.

- iv) If a cast jig or fixture drops down, it will probably break. It is not likely to bend and get out of alignment so as to result in the production of defective pieces.

ADVANTAGES OF FABRICATION METHOD.

Fabrication has the following advantages

- i. Standard parts can be used to build up the body.

- ii The jig or fixture can be built up quickly by using standard parts.

- iii After use, the jig or fixture can be disassembled.

ADVANTAGES OF WELDING METHOD.

- i The jig or fixture can be constructed speedily

- ii The welded construction is cheaper

- iii Less machining is needed than for fabricated parts.

COMPARISON

Castings is the most popular method. It is costly both in time and money but it has the property of absorbing and damping out vibrations, a very useful characteristic for milling fixtures. Castings are usually heavy and where lightness is needed, welded or fabricated jig or fixture has the advantage. A cast construction will break rather bend.

The welded construction must be stress relieved by heat treatment, to relieve all internal stresses induced during welding. Otherwise, the stresses will gradually distort the jig or fixture, as they relieve themselves. Only after stress relief can finish machining be done on the construction. If accuracy is to be held. And since machining must be done after assembly it is sometimes difficult to machine an internal surface of a welded construction. If welded jig or fixture gets dropped, it will get distorted. The effect will remain unnoticed until the device is used again when it produces defective parts or piece.

In a fabricated construction, all the component parts can be completely machined before assembly. This is often an important enough advantage to offset the lesser cost of the welded construction. Often a combination of the two methods is successfully used. In the fabrication method, the construction is held together by screws and dowels. The screws serve only to hold the part together with the dowels assuring accurate alignment.

Two dowels are sufficient in any case, whereas the screws can be one or more as needed. A fabricated construction is easy to repair, because disassembly is comparatively easy. To facilitate disassembly and assembly, each dowel should be a press fit in one part and slip in the other.

2.0

LOCATING AND CLAMPING.

The question of properly locating, supporting, and clamping the work is important, since the overall accuracy is dependent primarily on the accuracy with which the workpiece is consistently located within the jig or fixture. There must be no movement of the work during machining. Locating refers to the establishment of a proper relationship between the workpiece and the jig or fixture.

The function of clamping is to exert a force to press the workpiece against the locating surfaces and hold it there against the action of cutting forces.

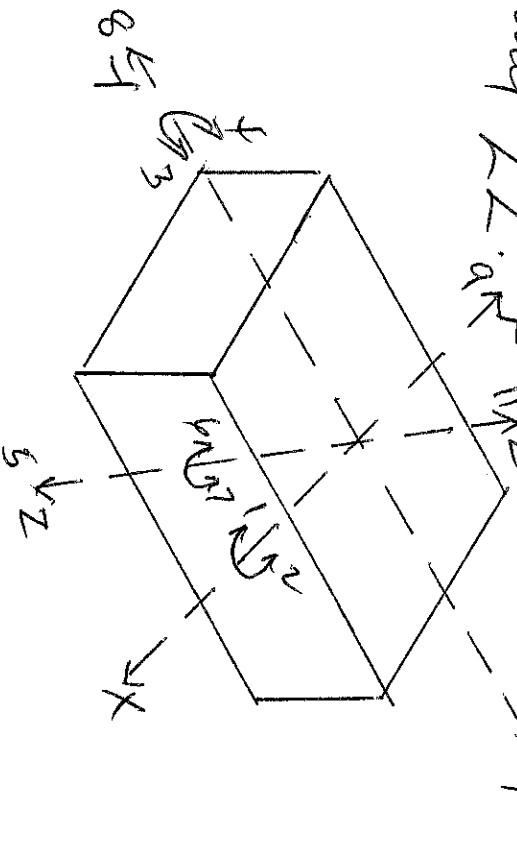
2.1

PRINCIPLE OF LOCATION.

In order to study the complete location of a workpiece within a jig or fixture, workpiece in space should be considered. The workpiece is assumed to have true and flat

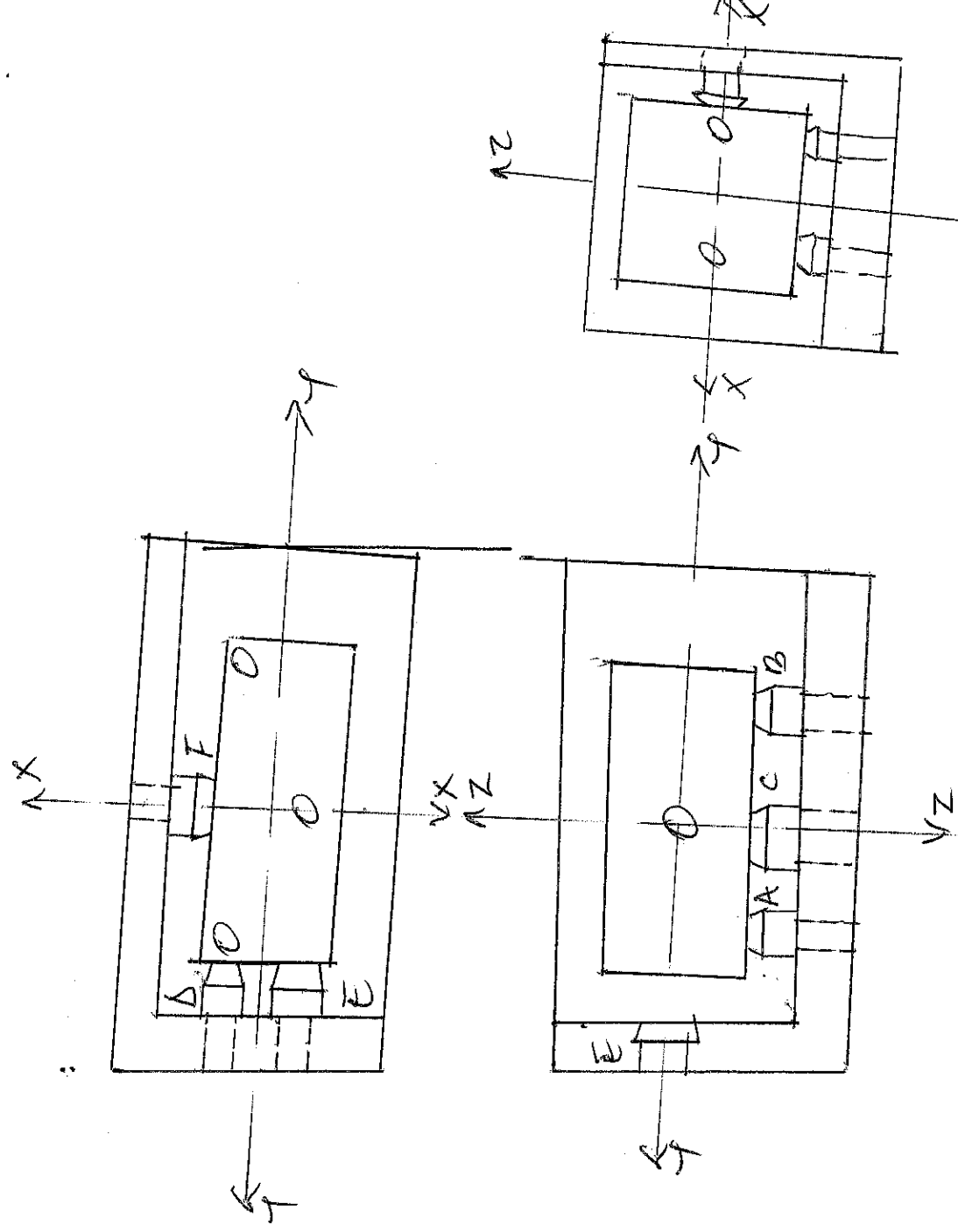
pg

faces. In a state of freedom, it may move in either of the two opposed directions along three mutually perpendicular axes X , Y , Z and YZ , XZ and XY .



Workpiece in space.

These six movements are called "movement of translation". Also, the workpiece can rotate in either of two opposed directions around each axis, clockwise and anti-clockwise. These six movements are called "rotational movements". The sum of these two types of movements gives the twelve degrees of freedom of a workpiece in space. To confine the workpiece accurately and positively in another fixed body (fig. a fixture), the movement of the workpiece in any of the twelve degrees of freedom must be restricted.



Workpiece located in a Fixed Body

- (a) The workpiece is resting on three pins A, B and C which are inserted in the base of the fixed body. The workpiece cannot rotate about the axes XX and YY and also it cannot move down ward. In this way, the five degrees of freedom 1, 2, 3, 4 and 5 have been arrested.
- (b) Two more pins D and E are inserted in the fixed body, in a plane perpendicular

to the Plane Containing the Pins A, B and C. Now the workpiece cannot rotate about the Z-axis and also it cannot move towards the left. Hence, the addition of pins A and C restrict three more degrees of freedom namely 6, 7 and 8.

① Another pin F in the second vertical face of the fixed body, arrest degree of freedom. Thus, six locating pins three in the base of the fixed body, two in a vertical plane and one in another vertical plane, the three planes being perpendicular to one another, restrict nine degrees of freedom. Three degrees of freedom namely 10, 11 and 12 are still free. To restrict these, three more pins, one for each of these degrees of freedom are needed. But this will completely enclose the workpiece making its loading and unloading into the jig or fixture impossible. Due to this these remaining three degrees of freedom may be arrested by means of a clamping device. The above method of locating a workpiece in a jig or a fixture is called a work-3-2-1" principle or "Six Point location" principle.

PRINCIPLES OF PIN LOCATION:

1. The Principle of Minimum Locating Points.

According to this principle, only the minimum locating points should be used to secure location of the workpiece in any one plane. Considering the "3-2-1" Principle, there pins are used in the base of the fixed body.

This is due to the reason that this is the minimum number of locating points through which a plane can be drawn on which the workpiece will seat. The workpiece may rock and get strained if more than three locating points are provided.

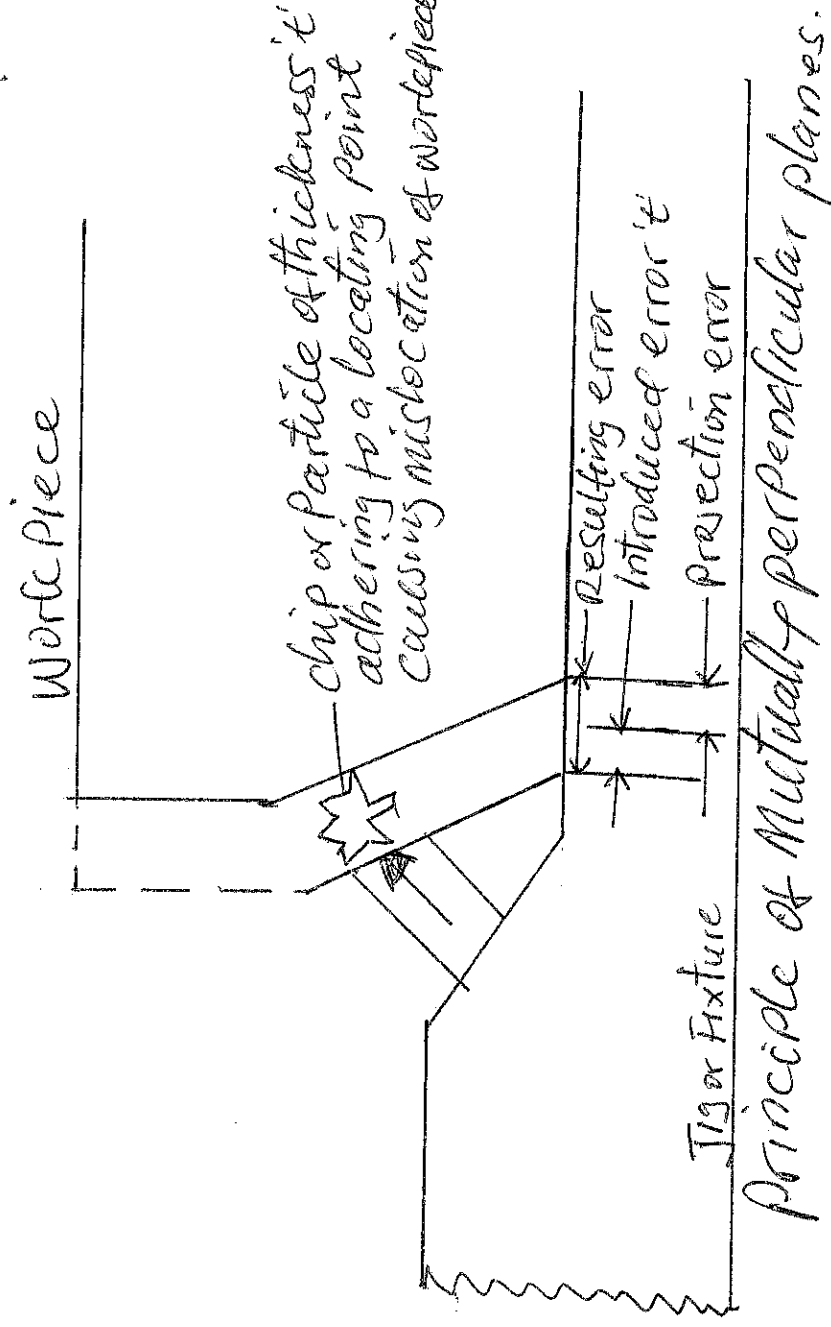
Now considering the second plane, it is clear that if one locating point is provided, the workpiece will swivel about this point, but not if there are two such locating points. These two locating points establish a line parallel to the first plane. With the workpiece located against a plane and a line, it has only one direction of movement in third plane. Therefore, one locating point is sufficient in the third plane to prevent this movement.

2. THE PRINCIPLE OF MUTUALLY PERPENDICULAR PLANES.

The "3-2-1" Principle can also be put as a Workpiece may be fully located by supporting it against three points in one plane, two points in second plane and one point in a third plane. These three planes are not parallel and are preferably perpendicular to one another - If the locating surfaces are not perpendicular to one another, the following two difficulties will arise.

(i) The Workpiece will tend to lift due to the wedging action between the two locating surfaces.

(ii) A large error in the movement of the Workpiece introduced due to the displacement of a locating point or a particle (chip or dirt) adhering to it, as is clear from the fig below. The difference of resulting error and introduced error, that is, the projection factor is zero when the locating ~~surfaces~~ are normal and increases as the angle between them becomes more acute.



- ③ THE PRINCIPLE OF EXTREME POSITIONS.
The locating points should be placed as far away from one another as possible, to achieve the greatest accuracy in location.